

Xinghao Qiao

Last update: September 3, 2025

CONTACT INFORMATION

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ACADEMIC APPOINTMENTS

The University of Hong Kong, Faculty of Business and Economics, Innovation and Information Management (IIM) area, **Business Analytics** group, Hong Kong

Associate Professor (without tenure) Dec., 2023 –

Tsinghua University, Center for Statistical Science, Beijing China

Visiting Associate Professor Apr. 2022 – Dec. 2022

London School of Economics, Department of Statistics, London U.K.

Associate Professor (**with tenure**) Aug. 2022 – Dec. 2023

Assistant Professor (tenure-track) Sep. 2015 – Jul. 2022

EDUCATION

University of Southern California, Marshall School of Business, Department of Data Sciences and Operations, Los Angeles, CA USA

Ph.D. in Business Statistics Aug. 2010 – Jul. 2015

- Advisor: Professor Gareth M. James

University of Chicago, Department of Statistics, Chicago, IL USA

M.S. in Statistics Sep. 2008 – Jun. 2010

Tsinghua University, Academic Talent Program, Beijing, China

B.S. in Mathematics and Physics Aug. 2004 – Jul. 2007

RESEARCH INTERESTS

Functional data analysis

Time series analysis

High-dimensional statistics/econometrics

AI and machine learning with business applications

PUBLICATIONS

*: Student co-author under my supervision; (✉): Correspondence author;

The authors' names are sorted *alphabetically* in all journal publications from 2022 to the present.

1. Li, D., Qiao, X.(✉) and Wang, Z.* (2025). Factor-guided estimation of large covariance matrix function with conditional functional sparsity, *Journal of Econometrics*, 251, 106070.
‘A’ journal in Economics at HKU
2. Guo, S., Li, D., Qiao, X.(✉) and Wang, Y.* (2025). From sparse to dense functional data in high dimensions: Revisiting phase transitions from a non-asymptotic perspective, *Journal of Machine Learning Research*, 26(15), 1-40.
‘A’ journal in Business Analytics at HKU
3. Guo, S., Qiao, X. (✉), Wang, Q.* and Wang, Z.* (2025). Factor modelling for high-dimensional functional time series, *Journal of Business & Economic Statistics*, in press.
4. Li, D. and Qiao, X. (2025). High-dimensional functional time series: A selective review, *Scientia Sinica: Mathematica*, in press.

5. Chang, J., Fang, Q.*, Qiao, X.(✉) and Yao, Q. (2024). On the modelling and prediction of high-dimensional functional time series, *Journal of the American Statistical Association*, in press.
‘A’ journal in Business Analytics at HKU
6. Fang, Q.*, Guo, S. and Qiao, X.(✉) (2024). Adaptive functional thresholding for sparse covariance function estimation in high dimensions, *Journal of the American Statistical Association*, 119, 1473-1485.
‘A’ journal in Business Analytics at HKU
7. Chang, J., Chen, C.*, Qiao, X.(✉) and Yao, Q. (2024). An autocovariance-based learning framework for high-dimensional functional time series, *Journal of Econometrics*, 239, 105385.
‘A’ journal in Economics at HKU
8. Liu, Y.*, Qiao, X.(✉), Pei, Y. and Wang, L. (2024). Deep functional factor models: Forecasting high-dimensional functional time series via Bayesian nonparametric factorization, *the 41th International Conference on Machine Learning (ICML)*, PMLR, 235, 31709-31727.
9. Liu, Y.*, Qiao, X.(✉), Wang, L. and Lam, J. (2023). EEGNN: Edge enhanced graph neural networks with a Bayesian nonparametric graph model, *the 26th International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR, 206, 2132-2146.
10. Guo, S. and Qiao, X.(✉) (2023). On consistency and sparsity for high-dimensional functional time series with application to autoregressions. *Bernoulli*, 29, 451-472.
11. Liu, Y.*, Qiao, X.(✉) and Lam, J. (2022). CATVI: Conditional and adaptively truncated variational inference for hierarchical Bayesian nonparametric models, *the 25th International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR, 151, 3647-3662.
12. Fang, Q.*, Guo, S. and Qiao, X.(✉) (2022). Finite sample theory for high-dimensional functional time series with applications. *Electronic Journal of Statistics*, 16, 527-591.
13. Chen, C.*, Guo, S. and Qiao, X. (2022). Functional linear regression: Dependence and error contamination. *Journal of Business & Economic Statistics*, 40, 444-457.
14. Lian, H., Qiao, X. and Zhang, W. (2021). Homogeneity pursuit in single index models based panel data analysis, *Journal of Business & Economic Statistics*, 39, 386-401.
15. Qiao, X.(✉), Qian, C.*, James, G. and Guo, S. (2020). Doubly functional graphical models in high dimensions, *Biometrika*, 107, 415-431.
‘A’ journal in Statistics at HKU and LSE
16. Qiao, X.(✉), Guo, S. and James, G. (2019). Functional graphical models. *Journal of the American Statistical Association*, 114, 211-222.
‘A’ journal in Business Analytics at HKU
17. Radchenko, P., Qiao, X. and James, G. (2015). Index models for sparsely sampled functional data. *Journal of the American Statistical Association*, 110, 824-836.
‘A’ journal in Business Analytics at HKU
18. Qiao, X., Wang, Z.*, Yao, Q. and Zhang, B. (2025). Weight-calibrated estimation for factor models of high-dimensional time series, under major revision at *Journal of the American Statistical Association*.
‘A’ journal in Business Analytics at HKU
19. Lan, T., Li, Z., Qiao, X., Li, J., Xu, H. Li, Z., Bai, L., Li, M., Tsung, F., Zhao, R. and Zhang, C. (2025). MultiFun-DAG: Multivariate functional directed acyclic graph based on causal decompo-

PAPERS UNDER
REVIEW

sition, under major revision at *INFORMS Journal on Computing* (among UTD 24 top business academic journals).

20. Fang, Q., Guo, S.(✉), Hong, Y.* and Qiao, X. (2025). On robust empirical likelihood for non-parametric regression with application to regression discontinuity designs, under major revision at *Journal of Econometrics*
‘A’ journal in Economics at HKU
 21. Fang, Q., Jiang, Q. and Qiao, X.(✉) (2025). Large-scale multiple testing for functional network models, submitted to *Journal of Machine Learning Research*.
‘A’ journal in Business Analytics at HKU
 22. Li, D., Qiao, X.(✉) and Yu, C.* (2025). Large covariance matrix estimation with factor-assisted variable clustering, submitted to *Journal of the American Statistical Association*.
‘A’ journal in Business Analytics at HKU
 23. Long, M.*, Qiao, X.(✉) and Li, Q. (2025). Functional knockoffs selection with applications to sparse functional models in high dimensions, submitted to *Statistica Sinica*.
- MANUSCRIPTS AVAILABLE
24. Li, B.*, Qiao, X., Wu, W. and Dette, H. (2025). Convergence of covariance and spectral density estimates for high-dimensional functional time series, to be submitted to *the Annals of Statistics*.
 25. Wang, Z.*, Qiao, X., Li, D. and Tong, H. (2025). On a new robust method of inference for general time series models, to be submitted to *Journal of Econometrics*.
 26. Wang, Z.*, Qiao, X., Li, D. and Tong, H. (2025). On robust inference for null recurrent time series.
 27. Shen, C*, Li, D., Liu, Y. and Qiao, X. (2025). Portfolio selection by GNN-based GARCH models.
 28. Chen, C.*, Qiao, X. and Wang, Z.* (2025). Inference for high-dimensional error-in-variables vector autoregression.
 29. Li, D., Qiao, X. and Wang, Z.* (2025). Factor modeling for matrix-variate functional and longitudinal time series.
 30. Chang, J., Jiang, Q., Qiao, X. and Yang, L.* (2025). Functional white noise test in high dimensions.
- WORKING PAPERS UNDER PREPARATION
31. Fang, Q., Guo, S., Han, Y. and Qiao, X. (2025+). Local or global? Adaptive thresholdings in dynamic covariance matrix estimation.
 32. Fang, Q., Qiao, X. and Wang, Z. (2025+). Time series Gaussian chain graph models: Identifiability and estimation.
 33. Du, C., Lin, Z., Meng, C. and Qiao, X. (2025+). KNN regression in Wasserstein space for multidimensional and nonlinear distributional chaotic systems.
 34. Guo, S., Li, D., Qiao, X. and Shen, C. (2025+). Error estimation in high-dimensional linear models with correlated errors.
 35. Dou, B., Guo, S., Qiao, X. and Yao, Q. (2025+). High-dimensional count time series: Poisson regression and factor modelling.
 36. Fang, Q., Qiao, X. and Wang, X. (2025+). Functional dynamic tensor decompositions with application to modeling and forecasting multi-country yield curves.

RESEARCH GRANTS	“Calibrating weights for improved estimation in factor models for high-dimensional time series”. HKU Seed Fund for PI Research - Basic Research	2025-2027
	“Modelling high-dimensional functional time series: Clustering and prediction”. HKU Seed Fund for Basic Research for New Staff	2024-2026
	“Feature learning for high-dimensional functional time series”. Partner Investigator. Australian Research Council Grant - DP230102250	2023-2026
	LSE Research Infrastructure and Investment Funds	2017-2020
	LSE Startup Research Grants	2015-2018

PHD SUPERVISION	1. Xinyu Wang, IIM at HKU (09/2024–current), HKU Presidential PhD Scholarship; master student under my supervision at LSE.	
	2. Zihan Wang, (joint with Dong Li), Statistics at Tsinghua University (09/2022–current).	
	3. Qin Fang, Statistics at LSE (01/2019–04/2022); Current: Assistant Professor of Business Analytics at University of Sydney Business School , Sydney, Australia.	
	4. Yirui Liu, Statistics at LSE (09/2017–07/2022); Current: Quantitative Researcher at JP Morgan, London, U.K.	
	5. Cheng Chen, Statistics at LSE (01/2017–07/2021); Current: Assistant Professor of Statistics at Southwestern University of Finance and Economics, Chengdu, China.	

TEACHING EXPERIENCE	The University of Hong Kong , Faculty of Business and Economics, Hong Kong	
	• CR Data Scientist Program: Forecasting and Predictive Analytics	June 2025
	• MSBA 7013A: Forecasting and Predictive Analytics	Module 3 2024-2025
	• IIMT 1640B & C: Probability and Statistics for Business	Spring 2024, 2025
	London School of Economics , Department of Statistics, London, U.K.	
	• Machine Learning: Practical Applications online certificate course	2019-2023
	• ST 300: Regression and Generalised Linear Models	MT 2016–2019, LT 2016–2019 ¹
	• ST 311: Artificial Intelligence	LT 2022-2023
	• ST 443: Machine Learning and Data Mining	MT 2017-2021, 2023, LT 2016-2017
	• ST 510: Foundations of Machine Learning (PhD core)	LT 2021-2023
	• Workshop for ST 425: Statistical Inference	MT 2015
	London Business School , London, U.K.	
	• PU239: Machine Learning for business PhD students	ST 2021
	University of Southern California , Marshall School of Business, Los Angeles, CA USA	
	• <i>Instructor</i>	Fall 2013
	BUAD 310 Applied Business Statistics	
	• <i>Teaching Assistant</i>	Spring 2012
	BUAD 310 Applied Business Statistics	

¹MT and LT correspond to Michaelmas Term and Lent Term, respectively.

HONORS AND AWARDS	Fellow of the Higher Education Academy (PR188060)	
	Nominated in LSE Students Union Teaching Awards, 2021	
	LSE Postgraduate Certificate in Higher Education, 2020 -	
	Nominated in LSE Students Union Inspirational Teaching Award, 2020	
	LSE Excellence in Education Award, 2018	
	Simons Fellowship to visit Issac Newton Institute at the University of Cambridge, 2018	
	Marshall PhD student Graduate Assistantship, 2010-2015	
ON CAMPUS SERVICES	Committee Chair for HKU 2025 Summer Workshop on Statistics and Data Analytics	2025
	Organizer for the Business Analytics and Statistics seminar series in IIM area	2024-2025
	Admission selector for MS in Business Analytics	2024-2025
	Teaching and learning coordinator of IIM area at FBE	2024-2025
	Faculty undergraduate committee	2024-2025
	Undergraduate staff-student consultative committee (UGSSCC)	2024-2025
	Organizing committee for HKU 2024 Summer Workshop on Statistics and Data Analytics	2024
	Co-organizer for LSE Statistics and Data Science seminar series	2023
	Departmental extended education committee member	2022-2023
	Departmental research committee member	2020-2022
	Affiliated member of the LSE Data Science Institute	2021-2023
	Admission selector for MSc in Data Science	2019-2021
	Organizer and academic supervisor for Postgraduate Practitioners' challenge	2017, 2021-2023
	Organizer and academic supervisor for Undergraduate Practitioners' challenge	2019
	Admission selector for BSc in Business Mathematics and Statistics.	2018-2019
	Co-organizer for LSE Statistics seminar series	2016-2018, 2020-2023
	Member of hiring panels for the posts of Assistant Professor in Data Science	2017-2019
	Member of hiring panel for the post of Professor in Data Science	2016
	Organizer for time series and statistical learning study group	2015-2018
	Admissions selector for MSc in Financial Statistics.	2015-2019
SEMINAR PRESENTATIONS	Invited seminar talk at the School of Data Science, Fudan University, Shanghai, China, Nov. 2025.	
	Invited seminar talk at the Department of Decision Analytics and Operations, City University of Hong Kong, Hong Kong, Nov. 2025.	
	Invited seminar talk at the Department of Statistics and Actuarial Science, University of Hong Kong, Hong Kong, Mar. 2023.	
	Invited seminar talk at the Area of Innovation and Information Management, University of Hong Kong Business School, Hong Kong, Mar. 2023.	
	Invited seminar talk at the Faculty of Economics, University of Cambridge, Cambridge, U.K., Feb. 2023.	
	Invited seminar talk at the Center for Statistical Science, Tsinghua Univeristy, Beijing, China, Dec. 2022.	
	Invited seminar talk at the School of Statistics, East China Normal University, Shanghai, China,	

Nov. 2022.

Invited seminar talk at Wang Yanan Institute for Studies in Economics, Xiamen University, Xiamen, China, Oct.2022.

Invited seminar talk at the School of Economics and Management, University of Chinese Academy of Science, Beijing, China, Jun. 2022.

Invited seminar talk at the Research School of Finance, Actuarial Studies and Statistics, Australian National University, Canberra, Australia, Mar. 2022.

Invited seminar talk at the Department of Statistics, Pennsylvania State University, Pennsylvania, USA, Dec. 2020.

Invited seminar talk at the School of Statistics, Southwestern University of Finance and Economics, Chengdu, China, Apr. 2019.

Invited seminar talk at the Theory and Algorithms in Data Science group, Alan Turing Institute, London, U.K., Feb. 2019.

Invited seminar talk at Theory of Machine Learning Study Group, London School of Economics, U.K., Jan. 2019.

Invited seminar talk at the Department of Mathematics, King's College London, London, U.K., Jan. 2019.

Invited seminar talk at the Center for Statistical Science, Tsinghua University, Beijing, China, Jan. 2019.

Invited seminar talk at the Isaac Newton Institute, University of Cambridge, Cambridge, U.K., May 2018.

Invited seminar talk at the High Dimensional Statistics study group, Alan Turing Institute, London, U.K., Mar. 2018.

Invited lectures at the School of Computer Science and Technology, Harbin Institute of Technology, Harbin, China, Jun. 2017.

Invited seminar talk at the Department of Mathematics, University of York, York, U.K., Nov. 2016.

Invited seminar talk at the School of Computer Science and Technology, Harbin Institute of Technology, Harbin, China, Jul. 2016.

Invited seminar talk at the S3RI-Southampton Statistical Sciences Research Institute, University of Southampton, Southampton, U.K., Oct. 2015.

Invited seminar talk at the Melbourne Business School, University of Melbourne, Melbourne, Australia, Mar. 2015.

Invited seminar talk at the Discipline of Business Analytics, University of Sydney Business School, Sydney, Australia, Mar. 2015.

Invited seminar talk at the Department of Statistics and Actuarial Science, the University of Hong Kong, Hong Kong, Mar. 2015.

Invited seminar talk at the Department of Information Systems, Business Statistics and Operations Management, Hong Kong University of Science and Technology, Hong Kong, Mar. 2015.

Invited seminar talk at the Department of Management Science, City University of Hong Kong, Hong Kong, Mar. 2015.

Invited seminar talk at the Department of Statistics, University of Pittsburgh, Pittsburgh, PA USA, Feb. 2015.

Invited seminar talk at the Department of Statistics, London School of Economics and Political Science, London, U.K., Feb. 2015.

Invited seminar talk at the Department of Statistics, Ohio State University, Columbus, OH USA,

CONFERENCE
PRESENTATIONS

Jan. 2015.

Invited seminar talk at the Data Sciences and Operations Department, Marshall School of Business, University of Southern California, Los Angeles, CA USA, Jan. 2015.

Invited seminar talk at the Department of Statistics, University of California-Los Angeles, Los Angeles, CA USA, Nov. 2014.

Invited talk at the 13th Conference of the IASC-ARS, Ho Chi Minh City, Vietnam, Dec. 2025.

Invited talk at the 3rd Joint Statistics and Data Science Meeting, Hangzhou, China, Jul. 2025.

Invited talk at the NUS/IMS Workshop: Frontiers of Functional Data Analysis: Challenges and Opportunities in the Era of AI, Singapore, Aug. 2024.

Invited talk at the 1st Joint Statistics and Data Science Meeting, Beijing, China, Jul. 2023.

Invited talk at Frontiers Forum on Data Science, Zhejiang University, Hangzhou, China, Aug. 2022.

Invited talk at Joint Statistical Meetings (JSM), Washington D.C., USA, August 2022.

Invited talk at IMS Annual Meeting, London, U.K., Jun. 2022.

Invited talk at 14th International Conference of the ERCIM WG on Computational and Methodological Statistics, London, U.K., Dec. 2021.

Invited talk at 13th International Conference of the ERCIM WG on Computational and Methodological Statistics, London, U.K., Dec. 2020.

Invited talk at 11th ICSA International Conference, Zhejiang University, Hangzhou, China, Dec. 2019.

Invited talk at LSE-SWUFE Workshop, Southwestern University of Finance and Economics, Chengdu, China, Dec. 2019.

Invited talk at 3rd International Conference on Econometrics and Statistics, National Chung Hsing University, Taichung, Taiwan, Jun. 2019.

Invited talk at 11th International Conference of the ERCIM WG on Computational and Methodological Statistics, Pisa, Italy, Dec. 2018.

Invited talk at the 5th Institute of Mathematical Statistics Asia Pacific Rim Meeting, Singapore, Jun. 2018.

Invited talk at Tsinghua Sanya International Workshop on “Complex Time Series Modelling and Forecasting: Dynamic Network, Spatial-temporal Data, and Functional Processes”, Sanya, China, Jan. 2018.

Invited talk at 10th International Conference of the ERCIM WG on Computational and Methodological Statistics, London, U.K., Dec. 2017.

Invited talk at Royal Statistical Society 2017 Annual Conference, Glasgow, U.K., Sep. 2017.

Invited talk at 2017 IMS China, Nanning, China, Jun. 2017.

Invited talk at 1st International Conference on Econometrics and Statistics, Hong Kong University of Science and Technology, Hong Kong, Jun. 2017.

Invited talk at 9th International Conference of the ERCIM WG on Computational and Methodological Statistics, Seville, Spain, Dec. 2016.

Invited talk at CRiSM workshop: Recent Developments in Large-scale Inference, University of Warwick, Coventry, U.K., Aug. 2016.

Invited talk at the 4th Institute of Mathematical Statistics Asia Pacific Rim Meeting, Chinese University of Hong Kong, Hong Kong, Jun. 2016.

Invited talk at 8th International Conference of the ERCIM WG on Computational and Methodological

ical Statistics, London, U.K., Dec. 2015.

Invited talk at WNAR/IMS annual meeting, Boise, Idaho USA, Jun. 2015.

Invited talk at joint meeting between IMS and Australian Statistical Conference, Sydney, Australia, Jul. 2014.

PROFESSIONAL SERVICES

Co-organizer for Tsinghua Sanya International Workshop on “Complex time series analysis: high-dimensionality, change-points, forecasting and causality”, Sanya, China, January 2024.

Scientific program committee for 2021 CMStatistics

Paper reviewers for *the Annals of Statistics*, *Journal of the Royal Statistical Society: Series B*, *Journal of the American Statistical Association: Theory & Methods*, *Journal of the American Statistical Association: Applications & Case Studies*, *Bernoulli*, *Biometrika*, *Journal of Econometrics*, *the Annals of Applied Statistics*, *Journal of Business and Economic Statistics*, *Journal of Machine Learning Research*, *Biometrics*, *Econometric Theory*, *Journal of Computational and Graphical Statistics*, *Journal of Multivariate Analysis*, *Statistica Sinica*, *Technometrics*, *Canadian Journal of Statistics*, *Scandinavian Journal of Statistics*, *Statistical Inference for Stochastic Process*, *Metrika*, *Journal of Applied Statistics*, *Statistics and Computing*, *Statistics in Medicine*, *Journal of Nonparametric Statistics*, *Mathematics*, *International Conference of Machine Learning*, *Conference on Neural Information Processing Systems*, *International Conference on Artificial Intelligence and Statistics*.

Chair and organizer for invited sessions, contributed topic sessions at many conferences including 2015 CMStatistics in London, 2016 IMS Asia Pacific Rim Meeting in Hong Kong, 2018 IMS Asia Pacific Rim Meeting in Hong Kong, 2018 International Conference in Econometrics and Statistics in Hong Kong, 2019 International Conference in Econometrics and Statistics in Taiwan, 2020 CMStatistics in London, 2021 CMStatistics in London, 2024 First Macau International Conference on Business Intelligence and Analytics.

STUDENTS

PhD thesis examination

Jianlong Shao, IIM, HKU (04/2024)

Philip Nadler, Data Science Institute, Imperial College London (10/2021),

Alexandra Tsimbalyuk, Statistics, LSE (05/2021)

DBA thesis examination at HKU FBE

Hao Qu (01/2025)

Fan Feng (05/2024)

Xiaoliang Chen (05/2024)

MSc in Business Analytics capstone project supervision at HKU

Luyu Chen, Yifan Du, Zhijing Huang, Hao Ma, Dingkan Peng, Haiqi Wu, Xuan Zhang, Yunya Zhang (2025) with *KC Group*.

Zhanpeng Chen, Jie Cheng, Siyu Huang, Nengruo Kuang, Jialu Xu, Qianqian Yu, Runqi Yu, Ailin Zhang (2024), with *KC Group*.

MSc (Research) supervision at LSE Statistics

Darren Yeap (2020–2021)

Hongshen Chen (2018–2019)

Yida Li (2018–2019)

Martin Pawelczyk (2017–2018)

Rahul Mahtani (2017–2018)

Srshti Putcha (2016–2017)

MSc in Data Science capstone project supervision at LSE

Zirui Liu, Jinlei Zeng, Jiajun Luo and Shuyuan Jia (2022–2023), with *Plymouth Marine Lab*.

Marta Rodriguez, Jessica Lambert and Anoushka Gupta(2021–2022), with *Deutsche Bank*.

Sishi Feng, Jingyi Wang and Jiayu Xia (2020–2021), with *AstraZeneca*.

Luis Alonso Ma, Kevin Beck and Yihuang Gu (2017–2018), with *LSE Timetable*.

Junyang Ge and Milton Lim (2017–2018), with *Virgin Atlantic*.

Malika Malik and Zhuomin Xiao (2017–2018), with *SpecSavers*.

Mengdi Xu and Yijin Zhao (2017–2018), with *Google*.